

Qatar-2b

R-1761

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_220310 · created 2026-08-02T20:59:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459648.9883 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.039
Transit depth (Rp/Rs)^2	7.6e-05 +/- 0.0068 [%]
Orbital Inclination (inc)	89.05 +/- 1.73
Airmass coefficient 1 (a1)	0.99 +/- 0.17
Airmass coefficient 2 (a2)	0.05 +/- 0.13
Scatter in the residuals of the lightcurve fit is	17.42 %
Best Comparison Star	None
Optimal Aperture	8.64
Optimal Annulus	18.01
Transit Duration (day)	0.0667 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.001 $\pm$ 0.039 vs 0.16327 (deviation 2.45 σ)
Duration fit vs published	0.0667 d vs 0.0767 d (deviation 2.1 σ)
Mid-time O-C	0.6 min
Depth SNR	0.0
Residual scatter	17.42 %

Light curve

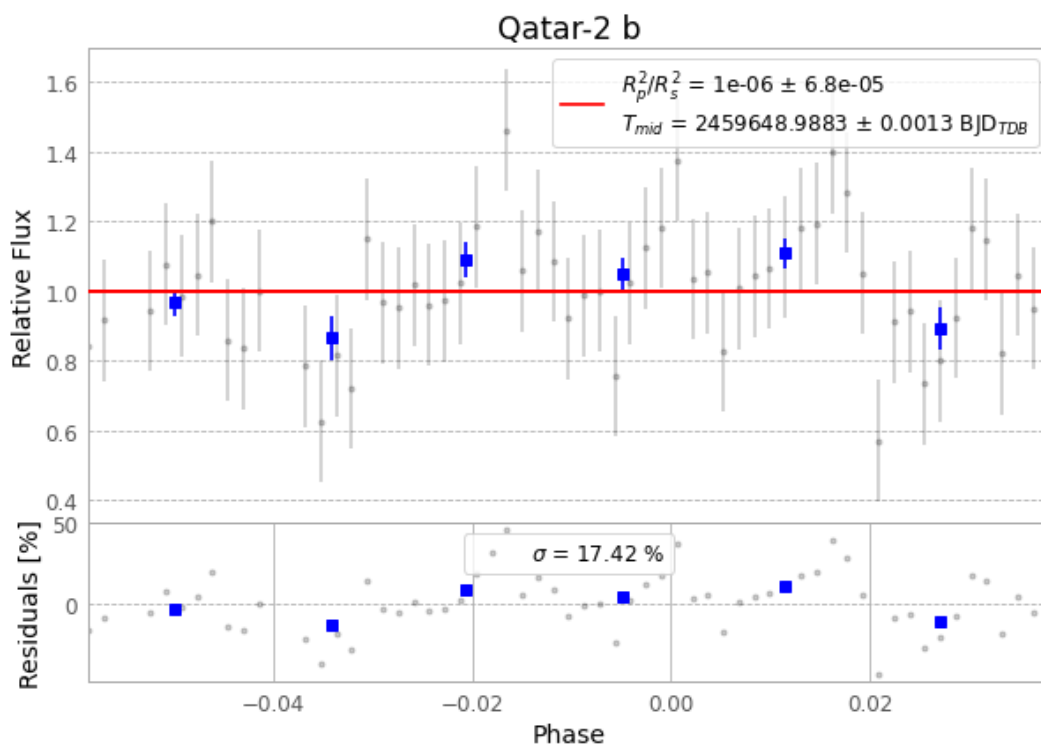


Figure 1 — FinalLightCurve_Qatar-2 b_2022-03-10.png (SHA-256 4e9358e2ca113eda...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [488, 121], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cae7a48e57a86c5c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), Qatar-2220310094212.FITS → Qatar-2220310124812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-220310.log (f1356651ee14...), FinalLightCurve_Qatar-2 b_2022-03-10.png (4e9358e2ca11...), FinalLightCurve_Qatar-2 b_2022-03-10.csv (56a12a33f349...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 20:59Z.